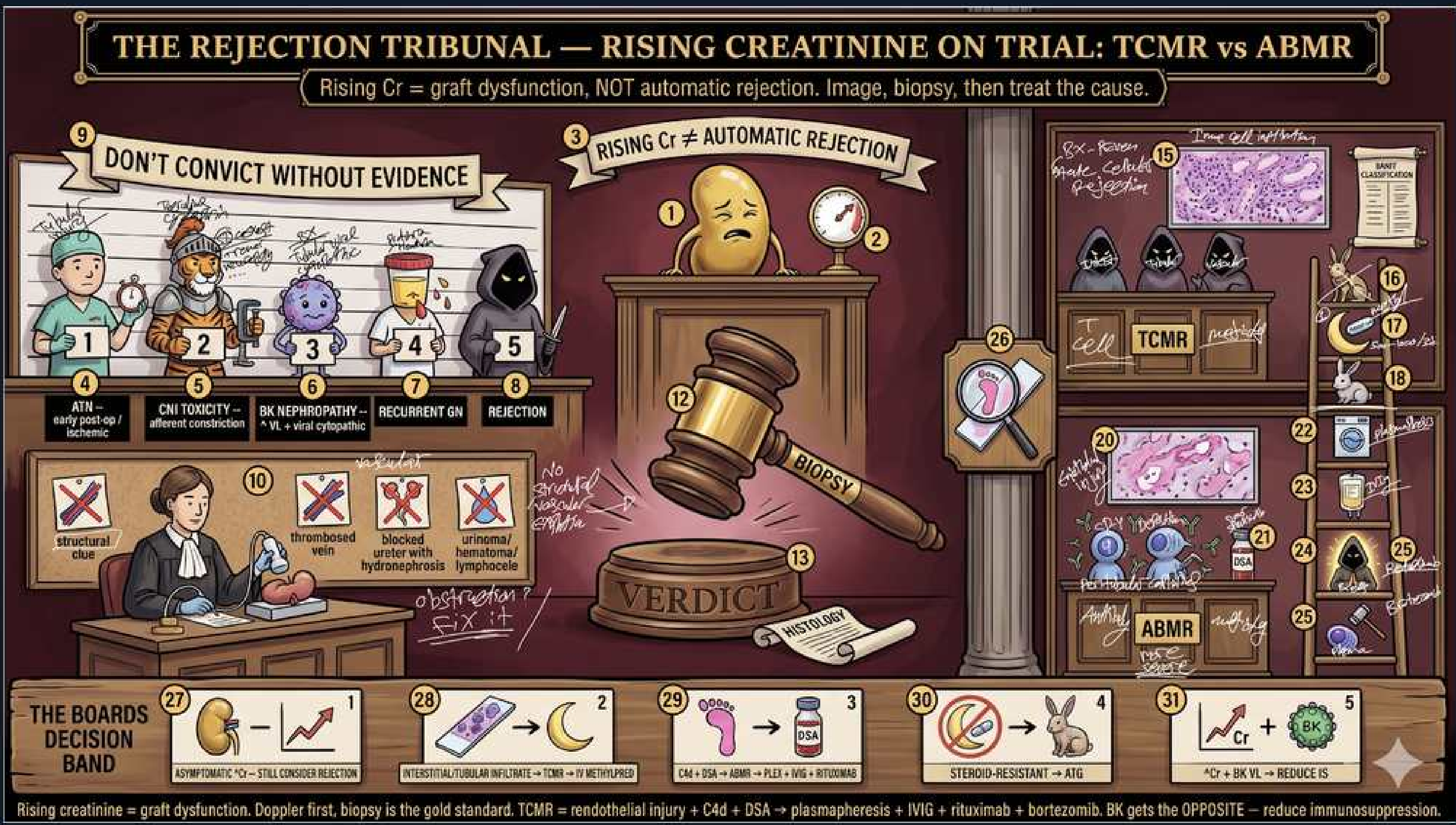
The Rejection Tribunal: Rising Creatinine, TCMR vs ABMR



1. Rising Cr ≠ automatic rejection

WHAT YOU SEE
Banner, potato/kidney with gauges

WHAT IT ENCODES
A rising creatinine signals graft dysfunction, NOT automatically rejection. The workup sequence is: rule out obstruction/volume, image (Doppler), then biopsy for diagnosis before treating. Empirically pulsing steroids without a diagnosis is wrong.

BOARD TRAP
Treating any Cr rise as rejection with steroid pulse — image and biopsy first; differential includes ATN, CNI toxicity, BK, RAS, recurrent GN, obstruction.

2. ATN (early post-op)

WHAT YOU SEE
Suspect #1 lineup, ATN early post-op ischemic

WHAT IT ENCODES
Delayed graft function from acute tubular necrosis is the most common early cause of a non-functioning graft, from ischemia-reperfusion injury (especially deceased-donor kidneys). It is self-limited; supportive dialysis bridges until recovery. Biopsy excludes rejection.

BOARD TRAP
Calling early delayed graft function 'rejection' — ATN/ischemia-reperfusion is the leading early cause and recovers with time, not immunosuppression.

3. CNI toxicity (afferent)

WHAT YOU SEE
Suspect #2, CNI toxicity afferent constriction

WHAT IT ENCODES
Calcineurin-inhibitor toxicity (tacrolimus/cyclosporine) causes afferent arteriolar vasoconstriction and a reversible Cr rise correlating with high trough levels. Lowering the dose improves function; chronic exposure causes striped fibrosis.

BOARD TRAP
Escalating immunosuppression for a Cr rise that is actually CNI toxicity — check trough levels; the fix is to LOWER the CNI dose.

4. BK nephropathy mimic

WHAT YOU SEE
Suspect #3, BK nephropathy, viral cytopathic

WHAT IT ENCODES
BK polyomavirus nephropathy mimics rejection on biopsy but is treated oppositely — by REDUCING immunosuppression. Diagnose with plasma BK PCR plus biopsy showing viral cytopathic tubular changes and decoy cells (SV40 stain).

BOARD TRAP
Increasing immunosuppression for biopsy inflammation that is BK nephropathy — that worsens it; BK requires REDUCING immunosuppression.

5. Recurrent GN

WHAT YOU SEE
Suspect #4, recurrent GN

WHAT IT ENCODES
The original glomerular disease can recur in the allograft (FSGS, IgA, MPGN, anti-GBM, lupus) and cause proteinuria and a rising Cr. Biopsy with immunofluorescence/EM distinguishes recurrence from rejection. FSGS recurrence can be early and aggressive.

BOARD TRAP
Attributing new proteinuria solely to rejection — recurrent glomerulonephritis (e.g., FSGS) is a distinct cause confirmed on biopsy.

6. Rejection (true)

WHAT YOU SEE
Suspect #5, grim reaper, rejection

WHAT IT ENCODES
True rejection is one suspect among many; it is confirmed by biopsy, not assumed. Rejection splits into T-cell mediated (TCMR) and antibody-mediated (ABMR), which look and are treated differently. Diagnosis drives therapy.

BOARD TRAP
Convicting rejection without biopsy — Banff-graded biopsy distinguishes TCMR from ABMR and from non-rejection mimics.

7. Biopsy is the gold standard

WHAT YOU SEE
Gavel labeled BIOPSY, judge

WHAT IT ENCODES
Allograft biopsy is the gold standard to diagnose the cause of graft dysfunction and to grade rejection (Banff classification). Doppler ultrasound first excludes obstruction and vascular causes; biopsy then delivers the verdict that guides treatment.

BOARD TRAP
Skipping biopsy and treating empirically — biopsy is the gold standard and the only way to separate TCMR, ABMR, and mimics.

8. TCMR: tubulointerstitial

WHAT YOU SEE
TCMR histology panel, tubulitis

WHAT IT ENCODES
T-cell mediated rejection shows tubulitis and interstitial inflammation (mononuclear infiltrate), sometimes endarteritis. Treatment is high-dose corticosteroid pulses; steroid-resistant cases get anti-thymocyte globulin (ATG). It responds to T-cell-directed therapy.

BOARD TRAP
Using plasmapheresis/IVIG for TCMR — that is ABMR therapy; TCMR is treated with steroid pulses and ATG if steroid-resistant.

9. ABMR: peritubular C4d

WHAT YOU SEE

ABMR histology panel, capillaritis

WHAT IT ENCODES

Antibody-mediated rejection shows microvascular inflammation (peritubular/glomerular capillaritis), C4d deposition in peritubular capillaries, and circulating donor-specific antibodies (DSA). Treatment: plasmapheresis + IVIG +/- rituximab/bortezomib. It is antibody-driven, not steroid-responsive alone.

BOARD TRAP

Relying on steroid pulses alone for ABMR — antibody-mediated rejection needs plasmapheresis, IVIG, and DSA removal, not just steroids.

10. C4d staining

WHAT YOU SEE

C4d positive peritubular capillaries

WHAT IT ENCODES

C4d deposition in peritubular capillaries is the histologic footprint of complement activation by donor-specific antibodies and a hallmark of antibody-mediated rejection. C4d positivity plus DSA plus microvascular injury defines ABMR on Banff criteria.

BOARD TRAP

Ignoring C4d as a non-specific stain — peritubular C4d is a key diagnostic marker pointing to antibody-mediated rejection.

11. Donor-specific antibodies

WHAT YOU SEE

DSA / antibody icon near ABMR

WHAT IT ENCODES

Circulating donor-specific antibodies (DSA) against HLA mediate ABMR; their presence with C4d and capillaritis confirms the diagnosis and predicts worse graft survival. Monitoring DSA helps assess response to plasmapheresis/IVIG.

BOARD TRAP

Diagnosing rejection by DSA alone — DSA must be paired with histologic injury (capillaritis, C4d) to call active ABMR.

12. TCMR therapy ladder

WHAT YOU SEE

Decision band, steroids → ATG

WHAT IT ENCODES

TCMR treatment ladder: pulse methylprednisolone first; if steroid-resistant, escalate to anti-thymocyte globulin (ATG) for severe or vascular rejection. The target is the alloreactive T-cell, not antibody.

BOARD TRAP

Jumping straight to ATG for mild TCMR — start with steroid pulses; reserve ATG for steroid-resistant or severe/vascular rejection.

13. ABMR therapy ladder

WHAT YOU SEE

Decision band, plasmapheresis/IVIG/rituximab

WHAT IT ENCODES

ABMR treatment combines plasmapheresis (remove DSA) + IVIG, often with rituximab (anti-CD20) and sometimes bortezomib (proteasome inhibitor to deplete plasma cells). Goal is to clear and suppress antibody production.

BOARD TRAP

Treating ABMR like TCMR with steroids/ATG alone — ABMR requires antibody removal (plasmapheresis + IVIG) and B/plasma-cell-directed agents.

14. BK gets the opposite

WHAT YOU SEE

Decision band, BK ↓ reduce IS

WHAT IT ENCODES

BK nephropathy is the great mimic that is treated the OPPOSITE way from rejection — by REDUCING immunosuppression, not adding it. This is why biopsy plus BK PCR is essential before escalating therapy for a rising Cr.

BOARD TRAP

Confusing BK with rejection and intensifying immunosuppression — BK is managed by reducing immunosuppression, the reverse of rejection.

15. Doppler / imaging first

WHAT YOU SEE

Magnifying glass, scope inspecting graft

WHAT IT ENCODES

Before biopsy, Doppler ultrasound excludes urinary obstruction, hydronephrosis, fluid collections, and vascular causes such as transplant renal artery/vein problems. Imaging is the cheap, fast first step that can spare an unnecessary biopsy.

BOARD TRAP

Biopsying before imaging — obstruction and vascular causes are found on ultrasound first and can fully explain a Cr rise.